🚀 Day 3: NumPy Basics – The Foundation of AI & Data Science

🎯 Day Goal

By the end of Day 3, you should understand:

- What NumPy is
- Why it's faster than Python lists
- Arrays
- Indexing
- Slicing
- Basic Operations

📖 What is NumPy?

NumPy (Numerical Python) is a Python library

used for numerical computing.
It provides a powerful data structure called ndarray (N-dimensional Array).
It is the foundation of:

- 📊 Data Science
- 🤖 Machine Learning
- 🧠 Deep Learning
- 📈 Data Analysis

🤔 Why NumPy Instead of Python Lists?

Python List	NumPy Array
Slower	Much Faster

More Memory	Less Memory
Limited Math Operations	Powerful Vectorized Operations

Example:

```
numbers = [1,2,3]
```

NumPy:

```
import numpy as np numbers = np.array([1,2,3])
```

📦 Creating Arrays

```
import numpy as np arr =
np.array([1,2,3,4,5]) print(arr)
```

Output

```
[1 2 3 4 5]
```

📏 Array Dimensions

1D Array

```
arr = np.array([1,2,3])
```

2D Array

```
arr = np.array([ [1,2,3], [4,5,6] ])
```

Looks like a table.

🎯 Indexing

Access an element using its position.

```
arr = np.array([10,20,30]) print(arr[0])
```

Output

```
10
```

Second element

```
print(arr[1])
```

Output

```
20
```

✂️ Slicing

Get multiple elements.

```
arr = np.array([10,20,30,40,50])
print(arr[1:4])
```

Output

```
[20 30 40]
```

Syntax

```
[start:end]
```

End index is not included.

➕ Basic Operations

Addition

```
arr = np.array([1,2,3]) print(arr+5)
```

Output

```
[6 7 8]
```

Multiplication

```
print(arr*2)
```

Output

```
[2 4 6]
```

Square

```
print(arr**2)
```

Output

```
[1 4 9]
```

Useful Functions

Length

```
len(arr)
```

Maximum

```
arr.max()
```

Minimum

Sum

```
arr.sum()
```

Mean

```
arr.mean()
```

💡 Real-Life Use Cases

NumPy is used for:

- Image Processing 📷
- AI Models 🤖
- Data Analysis 📊

- Scientific Computing 🔬
- Computer Vision 👁️
- Neural Networks 🧠

📝 Practice Questions

Easy

1. Create an array of numbers from 1 to 5.
2. Print the third element.
3. Multiply every element by 10.

Medium

Create

```
[[1,2], [3,4]]
```

Then print only:

```
4
```

Challenge

Create

```
[10,20,30,40,50]
```

Print

```
[20 30 40]
```

without manually typing the output.

🎯 Interview Questions

Q1. What is NumPy?

Answer: NumPy is a Python library used for fast numerical computing. It provides the ndarray data structure, which is faster and more memory-efficient than Python lists.

Answer: NumPy stores elements of the same data type in contiguous memory and performs operations using optimized C code, making computations significantly faster.

Q3. What is an ndarray?

Answer: An ndarray is NumPy's core data structure used to store homogeneous (same data type) multi-dimensional arrays.

Q4. What is the difference between Indexing and Slicing?

- Indexing retrieves a single element.
- Slicing retrieves a range of elements.

Example:

```
arr[2] # Indexing arr[1:4] # Slicing
```

🚀 Day 3 Checklist

- Install NumPy (pip install numpy)
- Create a 1D array
- Create a 2D array
- Practice indexing
- Practice slicing
- Perform arithmetic operations
- Solve the practice questions
- Commit today's code to GitHub

📌 Pro Tip

Many beginners think Pandas is the starting point for AI. In reality, NumPy is the mathematical foundation that libraries like Pandas, Scikit-learn, TensorFlow, and PyTorch build upon. Spending one solid day understanding NumPy basics will make the rest of your AI journey much smoother.